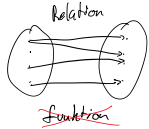


A binary relation from A to $B \subseteq A \times B$



funktion

Grafen for f er $\{(a,b) \in A \times B : b=f(a)\} \subseteq A \times B$

Ek. " $\leq$ " er en relation på $\mathbb{Z}$

$$R_{\leq} \stackrel{\text{def}}{=} \{(a,b) \in \mathbb{Z} \times \mathbb{Z} : a \leq b\}$$

Reflektiv

$$(a,a) \in R : \forall a \in A$$

der findes element eller

Symmetric if

$$(a,b) \in R \Rightarrow (b,a) \in R$$

Antisymmetric if

$$(a,b) \in R \Rightarrow (b,a) \notin R$$

Transitive if

$$((a,b) \in R \wedge (b,c) \in R) \Rightarrow (a,c) \in R$$

	P	<	≤	=	≠	q6
Ref	✓	✓	✓	✓	✓	✓
Iref	✓	✓	✓	✓	✓	✓
Sym	✓	✓	✓	✓	✓	✓
Asym	✓	✓	✓	✓	✓	✓
Trans	✓	✓	✓	✓	✓	✓

$P = \text{symmetric}$

$$s(R) = R \cup \{(b,a) \mid (a,b) \in R\}$$

Ek. $\leq$ er en "mindre end" på $\mathbb{Z}$

$$s(R) = \{(a,b) \in \mathbb{Z} \times \mathbb{Z} : a < b\} \cup \{(b,a) \mid (a,b) \in R\}$$

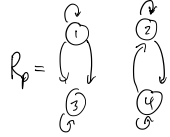
Representation of relations

$$A = \{1,2,3\} \text{ og } R_{\leq} = \{(1,2), (1,3), (2,3)\} \subseteq A \times A$$

Graf: Matrix: (adjacency matrix)

$$R_{\leq} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A = \{1,2,3,4\} \text{ og } R_{\text{paritet}} \stackrel{\text{def}}{=} \{(a,b) \in A \times A : a \text{ og } b \text{ har samme paritet}\}$$



$$R_p = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Notation

$$a \sim b$$

$$(aRb)$$

$$[a]_R = [b]_R$$

The Lexicographic Order

Eksempel $(A_1, \leq_1)$ og $(A_2, \leq_2)$

Så har vi $(A_1 \times A_2, \leq)$

hvor $(a_1, a_2) \leq (b_1, b_2)$

hvis $a_1 < b_1$

eller $a_1 = b_1$ og $a_2 \leq b_2$

Partielle ordninger

Reflektiv

Antisymmetrisk

Transitiv

(Partially ordered set)
Poset

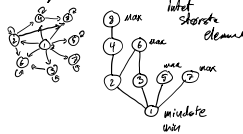
$$a \leq b \quad (a \neq b)$$

Eksempel	Samme paritet	<	≤	=	≠	≤	≤
Paritet	✓	✓	✓	✓	✓	✓	✓
Total	✓	✓	✓	✓	✓	✓	✓

Hasse Diagrams

Betrakt / på $\{1, \dots, 8\}$

min if $\neg \exists b \in$



Hasse diagram for en total ordning
er en lige linje